

**UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY
2026 INVITATIONAL B**

Biology

B01. C
B02. E
B03. B
B04. A
B05. C
B06. D
B07. A
B08. B
B09. C
B10. E
B11. A
B12. E
B13. D
B14. A
B15. B
B16. B
B17. A
B18. E
B19. B
B20. C

Chemistry

C01. D
C02. B
C03. E
C04. B
C05. C
C06. D
C07. A
C08. C
C09. A
C10. B
C11. E
C12. C
C13. D
C14. A
C15. A
C16. C
C17. A
C18. E
C19. C
C20. B

Physics

P01. E
P02. C
P03. D
P04. E
P05. D
P06. B
P07. B
P08. A
P09. C
P10. D
P11. B
P12. A
P13. D
P14. C
P15. C
P16. E
P17. B
P18. A
P19. E
P20. A

CHEMISTRY SOLUTIONS – UIL INVITATIONAL B 2026

- C01. (D) Potassium carbonate is K_2CO_3 , so there are six atoms.
- C02. (B) The balanced equation is $4 \text{Fe} + 3 \text{O}_2 \rightarrow 2 \text{Fe}_2\text{O}_3$
- C03. (E) Atomic mass is a weighted average of all naturally-occurring isotopes of an element and is the number at the bottom of a periodic table entry. Mass number is the mass of a single atom or a single isotope and is the sum of the protons and neutrons in the nucleus of that atom or isotope.
- C04. (B) A molecule has to have at least one double or triple bond to have π bonds. CH_4 , H_2O , and NH_3 all have only single bonds, while CO_2 has two double bonds and CO has one triple bond.
- C05. (C) $22.4 \text{ g CO}_2 / 44.01 \text{ g/mol} = 0.50898 \text{ mol CO}_2$.
 $PV = nRT$, $V = nRT/P = (0.50898)(0.08206)(298)/(1) = 12.4 \text{ L}$.
- C06. (D) The Lego SCUBA diver is made of ABS plastic which according to the data page has a density of $1.03 - 1.06 \text{ g/cm}^3$. This is greater than the density of water but less than the density of corn syrup, so the diver will sink to the bottom of the water (layer D) but float on the corn syrup (layer E).
- C07. (A) A positive ΔH means a reaction is endothermic and must absorb energy from its surroundings in order for the reaction to proceed.
- C08. (C) An atom is oxidized when its formal charge increases (becomes more positive). In reaction (i), the formal charge on the carbon atom changes from -4 to $+4$, in (ii) it also goes from -4 to $+4$, in (iii) it goes from $+2$ to $+4$, and in (iv) it goes from $+4$ to $+4$ (remains the same).
- C09. (A)
- C10. (B) Only raising the temperature will change K . The other changes will change the amounts of reactant and product present at equilibrium but will not change K itself.
- C11. (E) The p function means “take the $\log_{10}$ of this thing and change the sign,” so for example pH is the negative log of $[\text{H}^+]$. To go the other way, from $\text{p}X$ to just X , $X = 10^{-\text{p}X}$. The p function is commonly used in science to make a set of unwieldy numbers that range between 0 and 1 (such as acid concentrations ranging from 1 M down to 10^{-14} M) fit into an easier numeric range to work with, such as 0-14.
- C12. (C) The Cl^- ion present in solution from the NaCl will inhibit the solubility of the added AgCl due to the common Cl^- ion present from the dissolved NaCl . The other answers are wrong because: A and B: the common ion effect only applies when you have two different species with a common ion, not when you simply have more of the same compound; D) in this case the added NaCl would dissolve and some of the AgCl would precipitate out of solution. This is an example of the common ion effect, but the question specifically asks about affecting the solubility of the *added salt*, and in this case, it is the solubility of the *dissolved salt* that would be affected.

- C13. (D) The integrated rate law for the reaction relates the concentration of one reactant and time. For example, for a zero order reaction is $[A] = [A]_0 - kt$. Where $[A]$ is the current concentration of A , $[A]_0$ is the starting concentration, k is the rate constant for the reaction, and t is time. First order and second order reactions have different integrated rate laws that also relate $[A]$ to $[A]_0$ and t .
- C14. (A) There are two atoms bonded together so this is a molecule, but there is only one type of atom present, so this is an element, not a compound. A compound has to have more than one type of atom in it. This is not a mixture of any kind because all of the particles in the sample are the same. This is a pure diatomic element like N_2 or O_2 .
- C15. (A) For a rigorous solution to this you would create a RICE table and solve the resulting quadratic equation for x , where $x = [OH^-]$. However, when the K_b is small (generally around 10^{-5} or smaller) and the concentration is high (around 0.001 M or higher), x is small compared to the concentration of the base C_b and the quadratic equation simplifies to

$$[OH^-] = \sqrt{C_b K_b}$$

Where C_b = the concentration of the base. As long as $[OH^-] \ll C_b$ (typically 5% or less), the simplified equation can be used. In this case $[OH^-] = (0.75 \text{ M} \times 4.75 \times 10^{-5})^{1/2} = (0.0000356)^{1/2} = 0.0059687 \text{ M} = 0.0060 \text{ M}$ Checking the assumption that $[OH^-] \ll C_b$, $(0.0060/0.75) \times 100 = 0.8\%$, so the simplified equation is good. $[H^+][OH^-] = 1 \times 10^{-14}$, so $[H^+] = 1 \times 10^{-14} / 0.006 = 1.67 \times 10^{-12}$ and $pH = -\log(1.67 \times 10^{-12}) = 11.78$. You do not need the volume of the solution to solve this problem, pH is based only on the H^+ concentration.

- C16. (C) Arrow C is the difference in energy between the reactants and products. If the products have more energy than the reactants, as is the case here, the reaction is endothermic. In an exothermic reaction the products have a lower energy than the reactants.
- C17. (A) A decomposition reaction has one reactant and more than one product. Most decomposition reactions require heat.
- C18. (E) Cu has a molar mass of 63.54 g/mol and Cl has a molar mass of 35.45 g/mol. There are two Cl atoms in $CuCl_2$, so the molar mass of the compound is 134.45. The percent copper is $(63.54/134.45) \times 100\% = 47.3\%$
- C19. (C) $2 NaBrO_3 + Ba(NO_3)_2 \rightarrow Ba(BrO_3)_2 + 2 NaNO_3$ The sodium and nitrate ions are spectators, so the net ionic equation is $Ba^{2+}(aq) + 2 BrO_3^-(aq) \rightarrow Ba(BrO_3)_2(s)$. However, K_{sp} expressions are always written for the dissolution of the solid precipitate, so even though this question asks about forming a precipitate, the K_{sp} expression is still based on the reverse reaction, $Ba(BrO_3)_2(s) \rightarrow Ba^{2+}(aq) + 2 BrO_3^-(aq)$. The equilibrium constant is products over reactants raised to their stoichiometric coefficients, leaving out the solid, so $K_{sp} = [Ba^{2+}][BrO_3^-]^2$. (You can write a K_{eq} expression for the forward reaction which would be answer D, but that K_{eq} would not be K_{sp} .)
- C20. (B) 40 minutes = 2400 seconds. $2 \text{ mA} = 2 \text{ C/second} \times 2400 \text{ sec} = 4800 \text{ C}$. $4800 \text{ C} / 96485 \text{ C/mol} = 0.0497 \text{ mol of electrons} = 0.0497 \text{ mol of Ag} = 0.050 \text{ mol Ag}$.

PHYSICS SOLUTIONS – UIL INVITATIONAL B 2026

- P01. (E) page 63: “Soddy promptly purchased twenty milligrams to use in Ramsay’s laboratory for experiments on radium emanation. Their first attempts to isolate the emanation failed dismally, but spectral tests showed a helium line in the mixture of gases that radium produced.... With this larger sample, Soddy and Ramsay obtained nearly all of helium’s visible spectrum.”
- P02. (C) pages 72-73: “Villard decided to apply the definitive test for charged particles. He sent radium rays through a magnetic field. The magnet broke the beam into two parts. One part was deflected like beta rays. The other part continued in a straight line, oblivious to the magnet. This second portion...traveled through nearly ten inches of air, an aluminum plate, and several pieces of paper....If these powerful rays were neither alpha nor beta rays, what could they be? ... To continue the naming pattern established by Rutherford, these rays were called gamma rays...”
- P03. (D) page 83: “In 1906 physicists Norman Campbell and Albert B. Wood claimed that potassium and rubidium, two metals very different from uranium, radium, and thorium, were weakly radioactive. Tests eliminated possibilities that the effects came from trace impurities.... The radioactivity of potassium and rubidium seemed genuine.” Note: radioactive potassium-40 is very common in nature. Your body contains some, as do all foods that are sources of potassium, such as bananas.
- P04. (E) The cooler the star, the more slowly it uses up its core hydrogen fuel. So, the longest-lived stars are the stars with the lowest temperatures – that is, the class M red stars. Class O stars use up their fuel quickly and remain on the main sequence for only a few million years, Our Sun is a class G yellow star whose lifetime on the main sequence is around 10 billion years. Class M red stars remain on the main sequence for hundreds of billions of years.
- P05. (D) First, let’s convert the distance into miles:

$$d = 120 \text{ yards} * \frac{3 \text{ feet}}{1 \text{ yard}} * \frac{0.3048 \text{ m}}{1 \text{ foot}} * \frac{1 \text{ mile}}{1609 \text{ m}} = 0.068196 \text{ miles.}$$
 Now, we convert the time to hours:

$$t = 12.5 \text{ min} * \frac{1 \text{ hour}}{60 \text{ min}} = 0.20833 \text{ hours.}$$
 Thus, the velocity in miles per hour is

$$v = \frac{d}{t} = \frac{0.068196 \text{ miles}}{0.20833 \text{ hour}} = 0.327 \text{ mph.}$$
- P06. (B) Since we know the time during which the walnut is falling, we can use the kinematic equation:

$$x = x_i + v_i t + \frac{1}{2} a t^2.$$
 The final position is the floor ($x = 0$) and taking downward to be negative, we get: $0 = 2.35 + v_i(0.250) + \frac{1}{2}(-9.80)(0.250)^2 = 2.35 + 0.25v_i - 0.30625.$ This leads to $-2.044 = 0.25v_i \rightarrow v_i = -8.18 \text{ m/s}.$ Thus, the initial velocity was $v_i = 8.18 \text{ m/s}$ downward.
- P07. (B) Since there is no friction, the only two forces acting on the notebook are gravity (mg , directed downward), and the normal force (F_N , directed up and right, perpendicular to the incline). As we usually do when on an incline, we tilt our coordinate system so that the positive x-direction points down and right, parallel to the inclined plane, and the positive y-direction points up and right, perpendicular to the inclined plane. This means that the normal force points entirely in the positive y-direction. The gravitational force is at an angle with respect to our axis system, but it can be broken into components. In the positive x-direction, we have $mg \sin \theta$, and in the negative y-direction we have $mg \cos \theta$. Here $\theta = 36.0^\circ$ is the angle of the incline with respect to horizontal. We have no motion in the y-direction, so all forces in the y-direction must sum to zero. That is $\sum F_y = 0 \rightarrow F_N - mg \cos \theta = 0 \rightarrow F_N = mg \cos \theta.$ Thus, we find that the magnitude of the normal force is $F_N = (1.50 \text{ kg})(9.80 \text{ m/s}^2) \cos(36.0^\circ) = 11.9 \text{ N}.$

- P08. (A) By conservation of momentum, the total momentum of the system prior to the collision must equal the total momentum of the system after the collision. Prior to the collision, the cart has no momentum since it is at rest. The only momentum is that of the clay blob:
 $p_i = m_{cl}v_{ci} = (0.350\text{kg})(13.0\text{m/s}) = 4.55\text{kgm/s}$. After the collision, both the cart and the clay are moving, and since they are stuck together, they must be moving at the same velocity. So, the final momentum is $p_f = m_{cl}v_f + m_{ca}v_f = (0.350)v_f + (0.600)v_f = 0.950v_f$. By conservation of momentum $p_i = p_f \rightarrow 4.55 = 0.950v_f \rightarrow v_f = 4.79\text{m/s}$.
- P09. (C) First, we convert the angular displacement into radians. The merry-go-round takes two complete revolutions to reach maximum velocity. Thus, $\Delta\theta = 2.0\text{rev} * \frac{2\pi\text{rad}}{1\text{rev}} = 4\pi\text{radians}$. We know nothing about the time it takes to reach the maximum velocity, so we'll use the rotational kinematic equation that does not include time: $\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$. This gives $(9.00)^2 = (0)^2 + 2\alpha(4\pi)$, and leads to $8\pi\alpha = 81$, resulting in an angular acceleration of $\alpha = \frac{81}{8\pi} = 3.22\text{rad/s}^2$.
- P10. (D) At 20.0°C , the speed of sound is 343m/s . Since we know the frequency and the speed of sound, we can determine the wavelength of the resonant sound by using $f\lambda = v$. This gives $(144\text{Hz})\lambda = 343$, which leads to a wavelength of $\lambda = 2.38\text{m}$. We relate this wavelength to the length of the log by $n\lambda = 2L$, which gives $n(2.38\text{m}) = 2(4.76) \rightarrow n = 4$.
- P11. (B) Based on the density and mass of the cube of aluminum, we can find its volume. $\rho = \frac{m}{V} \rightarrow 2700\text{kg/m}^3 = \frac{2.80\text{kg}}{V} \rightarrow V = 0.001037\text{m}^3$. Now we can determine the buoyant force acting on the cube: $F_B = \rho_{\text{liquid}}Vg = (1000\text{kg/m}^3)(0.001037\text{m}^3)(9.80\text{m/s}^2) = 10.16\text{N}$. The weight of the cube in air is $W_{\text{air}} = mg = (2.80\text{kg})(9.80\text{m/s}^2) = 27.44\text{N}$. Thus, the apparent weight of the cube of aluminum while submerged in water is $W_{\text{water}} = W_{\text{air}} - F_B = 27.44 - 10.16 = 17.3\text{N}$.
- P12. (A) First, the resistance of the pasta can be found from Ohm's Law:
 $V = IR \rightarrow 24.0\text{V} = (56.0 \times 10^{-9}\text{A})R \rightarrow R = 4.286 \times 10^8\ \Omega$. Now we find the cross-sectional area of the strand of pasta: $A = \pi r^2 = \pi \left(\frac{D}{2}\right)^2 = \pi \left(\frac{1.20 \times 10^{-3}}{2}\right)^2 = 1.131 \times 10^{-6}\text{m}^2$. Finally, we relate the resistance of the pasta to its resistivity: $R = \frac{\rho L}{A} \rightarrow 4.286 \times 10^8 = \frac{\rho(0.360\text{m})}{1.131 \times 10^{-6}\text{m}^2} \rightarrow \rho = 1346 \approx 1350\ \Omega\text{m}$
- P13. (D) Electric potential is, thankfully, not a vector. Thus, we only need to consider the distance of each charge from the origin. The relevant equation for electric potential is $V = \frac{kQ}{r}$ where r is the distance from the charge to the point of interest. For the first charge, which is 400.0nC , the distance from the origin is 24.0cm . So, the potential due to the first charge is $V_1 = \frac{(8.99 \times 10^9)(400.0 \times 10^{-9})}{0.240} = 14980\text{V}$. For the second charge, which is 250.0nC , the distance from the origin is 18.0cm . Thus, the potential due to the second charge is $V_2 = \frac{(8.99 \times 10^9)(250.0 \times 10^{-9})}{0.180} = 12490\text{V}$. Now, we simply add the electric potentials due to each charge to get the total electric potential at the origin:
 $V_{\text{total}} = 14980 + 12490 = 27470\text{V} \approx 27500\text{V} = 27.5\text{kV}$.
- P14. (C) The radius of the circle traced out by a charged particle in a magnetic field is given by $r = \frac{mv}{qB}$. The field is perpendicular to the motion of the particle, and the charge of the particle is the same as a proton ($q = 1.602 \times 10^{-19}\text{C}$). Thus, we have: $2.20 = \frac{m(4.50 \times 10^6)}{(1.602 \times 10^{-19})(0.780)}$. This gives a mass of $m = 6.109 \times 10^{-26}$. Compared to the mass of a proton, this particle's mass is:
 $m = \frac{6.109 \times 10^{-26}}{1.67265 \times 10^{-27}} m_p = 36.5 m_p$.

- P15. (C) When an unpolarized beam of light passes through a polarizer, the beam's intensity is reduced by half due to the elimination of its cross-polarized components. Thus, after the first polarizer, we have $I_1 = \frac{1}{2}I_0 = \frac{1}{2}(600) = 300 \text{ W/m}^2$. The beam is now horizontally polarized. To get through the second polarizer, we rely on Malus' Law: $I_2 = I_1(\cos \theta)^2$ where θ is the angle between the polarization of the beam and the polarizer. In this case $\theta = 30.0^\circ$. So, the intensity after going through the second polarizer is $I_2 = (300)(\cos(30.0))^2 = 225 \text{ W/m}^2$.
- P16. (E) The equation that describes the location of an image created by a refracting surface is $\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_2 - n_1}{R}$. Because the surface of the water is flat, the radius of curvature is effectively infinite. That is $R = \infty$. Thus, the right-side term $\frac{n_2 - n_1}{R} = \frac{n_2 - n_1}{\infty} = 0$. So, for a flat surface we have: $\frac{n_1}{p} + \frac{n_2}{q} = 0$. The object (the minnow) is in the water, so $n_1 = 1.33$ and we see an image from a vantage point in the air, so $n_2 = 1.00$. The image is what we see, which is located 12.0cm below the water's surface. Since it is below the surface, the image location is negative. Thus, $q = -12.0\text{cm}$. Putting it all together, we can find the object location (where the minnow actually is located): $\frac{n_1}{p} + \frac{n_2}{q} = \frac{1.33}{p} + \frac{1.00}{-12.0} = 0 \rightarrow \frac{1.33}{p} = \frac{1}{12}$. This gives an object location of $p = 1.33(12.0) = 16.0\text{cm}$ below the surface.
- P17. (B) First, let's consider the Rydberg formula, which relates spectral line wavelengths to Hydrogen's atomic energy levels: $\frac{1}{\lambda} = R_H \left(\frac{1}{n_B^2} - \frac{1}{n_A^2} \right)$. Here $R_H = 1.0974 \times 10^7 \text{ m}^{-1}$ is the Rydberg constant, n_A is the higher energy level involved and n_B is the lower energy level involved. Putting in the wavelength gives us: $\frac{1}{1005 \times 10^{-9}} = (1.0974 \times 10^7) \left(\frac{1}{n_B^2} - \frac{1}{n_A^2} \right) \rightarrow \frac{1}{n_B^2} - \frac{1}{n_A^2} = 0.0907$. We don't have any more information from the problem itself, but the answer choices all end at either the $n_B = 3$ or the $n_B = 4$ levels. Let's try them both and see what we get for the starting level. For $n_B = 4$ we get $\frac{1}{16} - \frac{1}{n_A^2} = 0.0907 \rightarrow n_A^2 = -35.5$, which gives an undefined answer. So, $n_B = 4$ cannot be the correct final state. Let's consider $n_B = 3$ instead: $\frac{1}{9} - \frac{1}{n_A^2} = 0.0907 \rightarrow n_A^2 = 49.0 \rightarrow n_A = 7$. Thus, we can see that this spectral line wavelength is the result of a transition from the $n_A = 7$ to the $n_B = 3$ energy levels.
- P18. (A) First, we should note that the "Higgs force" isn't a real force. There are only four fundamental forces, listed as choices A through D. So, we can immediately eliminate E as a choice. Second, the gravitational force has not been observed on the subatomic level. It is far too weak and has not yet been connected to any type of particle decay. This eliminates D as a choice. Now we consider the properties of the K^+ meson. Notice that it has strangeness and a relatively long lifetime. The decay products are pions, which are composed of up and down quarks, and (importantly) have zero strangeness. Thus, a quark flavor change must have taken place in order to go from a strange particle to decay products without strangeness. This suggests that the Weak force is involved. The lifetime confirms it. Strong force decays are very fast ($\sim 10^{-23}\text{s}$) and even Electromagnetic force decays are reasonably fast ($\sim 10^{-16}\text{s}$). Weak force decays are the only ones that are as slow as 10^{-8}s . Thus, the Weak force is primarily responsible for the decay of the K^+ meson.

- P19. (E) The important equation that relates the stretching of a spring to the force pulling on the spring is Hooke's Law: $F = -k\Delta x$. The sign indicates that the spring force is opposite to the direction of the stretching, but we don't need to consider directions to solve this problem, so we can ignore it. Noting that the force to stretch the spring is provided by the weight of each mass, we get: $F = mg$. Putting it all together (and removing the unneeded negative sign), we get: $mg = k\Delta x = k(x - x_0)$, where x_0 is the unstretched length of the spring. The plot is of length versus mass, or x versus m . Rearranging, we see: $\frac{g}{k}m = x - x_0 \rightarrow x = \frac{g}{k}m + x_0$. Now we can tell that the slope of the line on the plot must equal $\frac{g}{k}$. To find the slope, we choose two points on the best fit line that are not data points. I shall choose (140g, 20cm) and (325g, 30cm). Converting to proper units gives us the points (0.140kg, 0.20m) and (0.325kg, 0.30m). Now we find the slope using these two points:
- $$slope = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0.30 - 0.20}{0.325 - 0.140} = \frac{0.10}{0.185} = 0.54 \text{m/kg.}$$
- Equating this to the slope formula derived earlier, we obtain: $0.54 = \frac{g}{k} \rightarrow k = \frac{9.80}{0.54} = 18.1 \approx 18 \text{N/m.}$
- P20. (A) The electric field is proportional to the gradient (rate of change over distance) of the potential. So, the more steeply the potential changes, the greater the electric field. On the equipotential map, steep changes in potential appear as equipotential lines that are closely spaced together. The most closely spaced lines are clearly at point A, which is where the electric field magnitude is the greatest. Point B represents a steep turning point of one equipotential, but that is not indicative of a large electric field. Points C and E are local maximums in potential, but they are not near areas of large gradients. Maximum potential does not equate to maximum electric field. Finally, point D is where the lines are farthest apart, representing the area of lowest electric field magnitude.